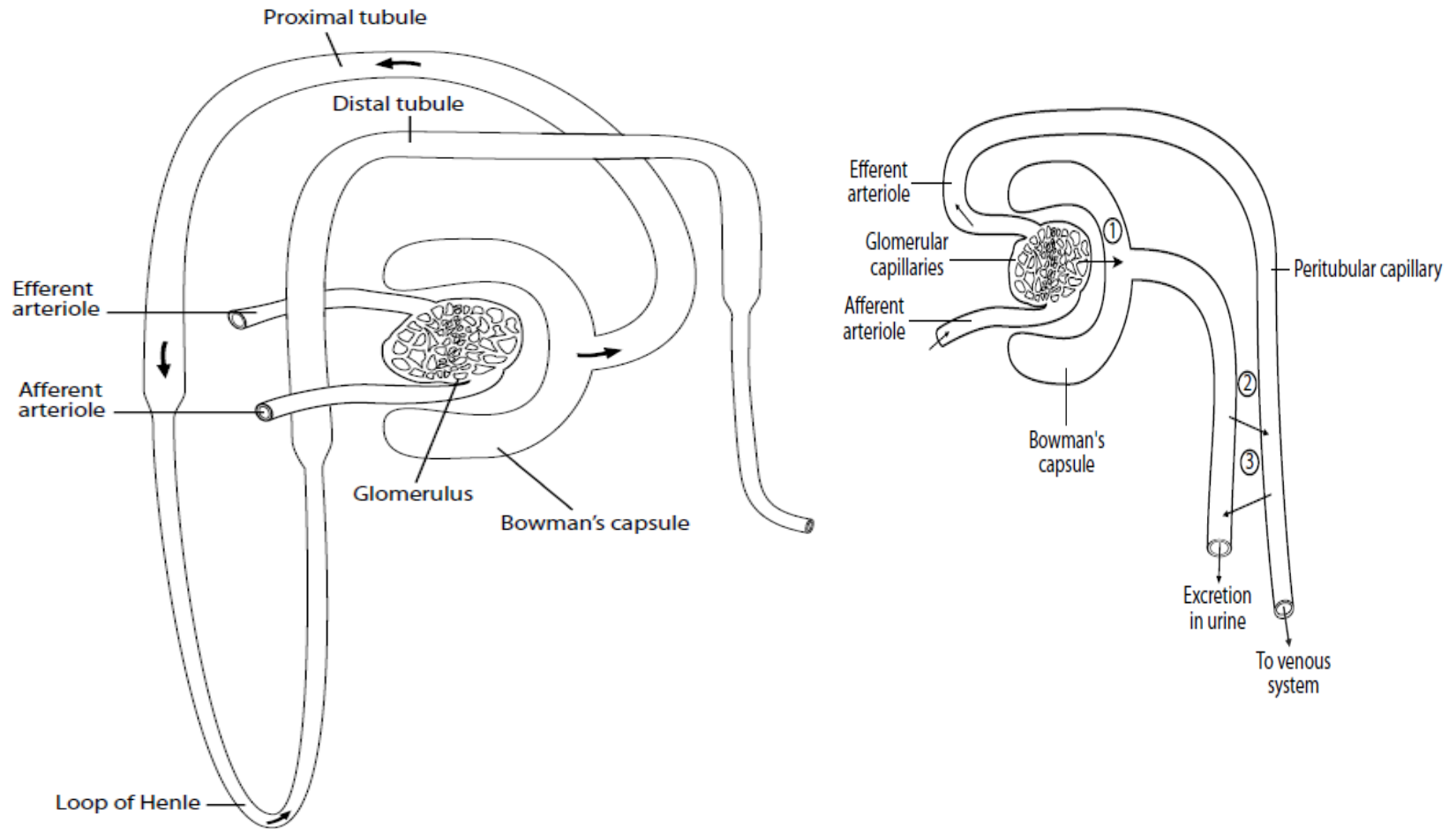
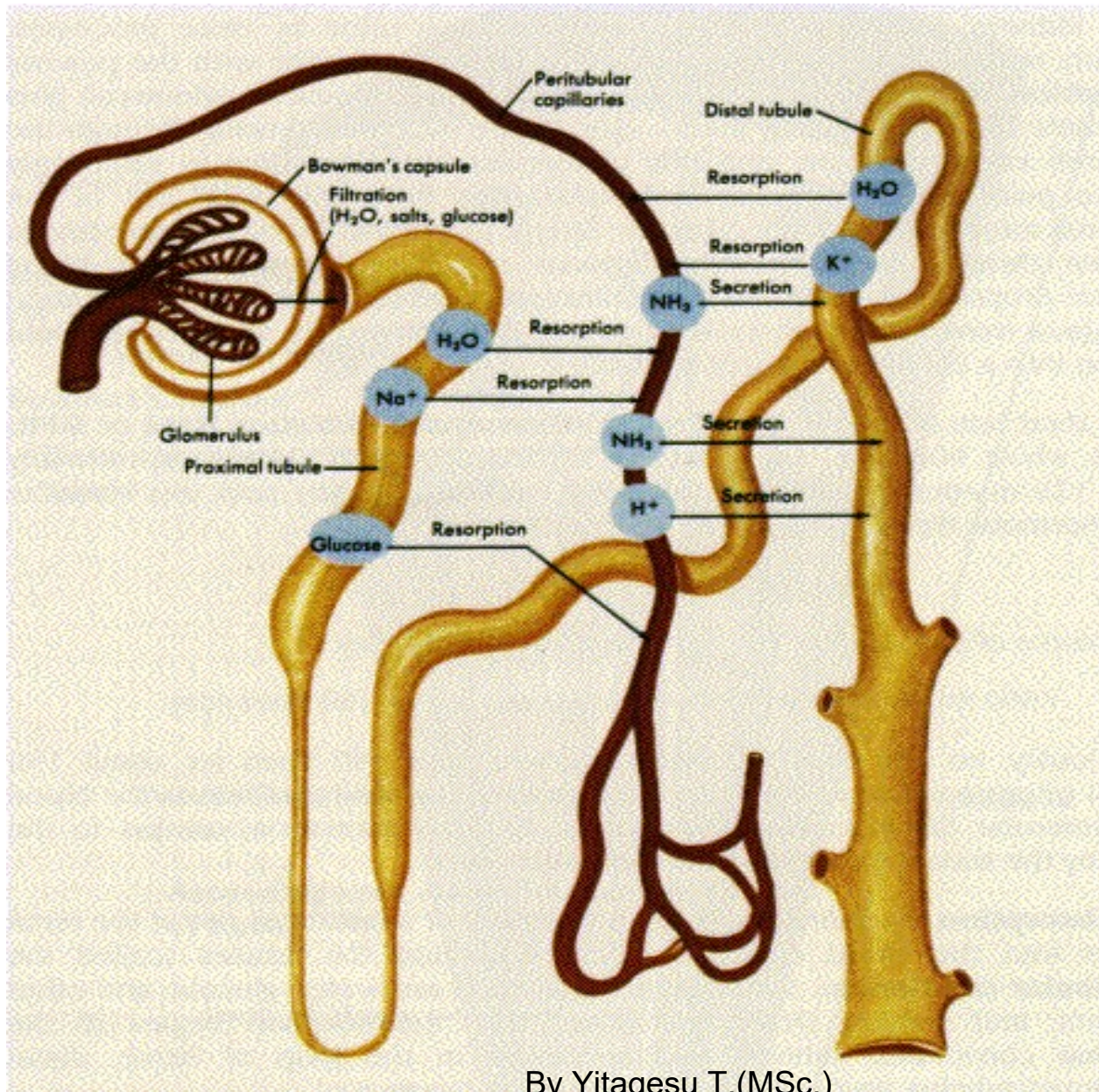


DIURETICS

Renal anatomy and Physiology





By Yitagesu T.(MSc.)

DIURETICS

- Diuretics are drugs that promote the output of urine excreted by kidney.
- Diuretics mainly promotes the excretion of the sodium ions(Na^+), chloride ions (Cl^-) or bicarbonate ions(HCO_3^-) and water from the body, the net result being increase the urine flow.
- These drugs act by decreasing tubular reabsorption
 - do not affect the glomerular filtration rate or the action of anti diuretic hormone (ADH)

DIURETICS...

- Diuretics are very effective
 - for the treatment of cardiac edema (CHF)
 - hyper tension
 - Nutritional edema
 - edema of pregnancy
 - cirrhosis of liver
 - lower the intracellular and cerebrospinal fluid pressure.

DIURETICS...

CLASSIFICATION

I. Carbonic anhydrase inhibitors (Site-I Diuretics—proximal tubule)

Acetazolamide, Methazolamide, Dichlorphenamide, Disulfamide, Ethoxzolamide.

II. Thiazide and Thiazide like Diuretics (Site-III Diuretics- distal tubule)

- Chlorthiazide, Benzthiazide, Hydrochlorothiazide

DIURETICS...

III. Loop Diuretics (Site-II Diuretics)

- Organo mercurials – Chlormerodine mercury, Meralluride, Mercaptomerin, Merethoxylline procaine, Mersalyl.
- 5-Sulpamoyl & 3-Amino Benzoic acid derivatives- Bumetanide, Furosemide,
- 4-Amino-3-pyridine sulphonyl ureas- Torsemide, Triflocin.
- Phenoxy acetic acids- Ethacrynic acid.

DIURETICS...

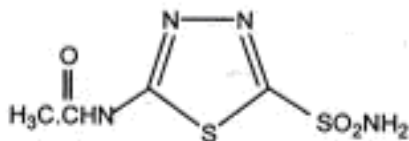
IV. Potassium sparing Diuretics (Site-IV Diuretics—cortical collecting duct)

1. Aldosteron inhibitors – Spiranolactone, Metyrapone
2. 2,4,7-Triamino-6-aryl pteridines – Triamterene
3. Pyrazinoyl Guanidines – Amiloride. HCl.

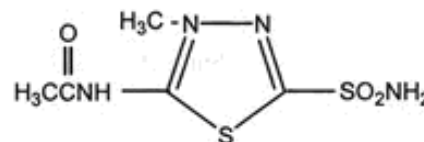
V. Miscellaneous -

Mannitol, Potassium acetate, Sodium acid phosphate, Urea, Caffeine, Theophylline. Theobromine.

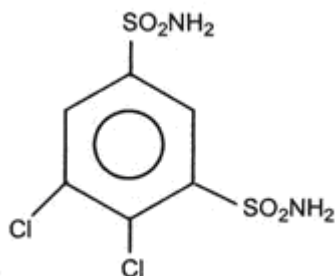
I. Carbonic anhydrase inhibitors (Site-I Diuretics)



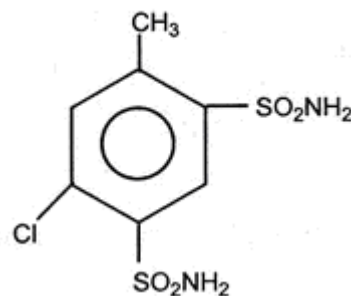
Acetazolamide



Methazolamide



diclofenamide



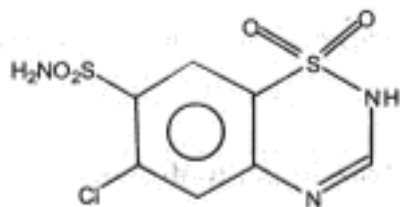
Disulfamide

DIURETICS...

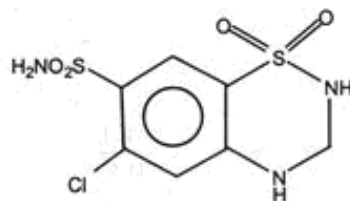
Structure Activity Relationship for CAI

- **The free sulfamoyl nitrogen** is important for diuretic activity.
- Substitution of the **methyl group** on one of the ring nitrogen (Methazolamide) retains the activity.
- The heterocyclic sulphonamides have **highest lipid/water partition coefficient** and **lowest pKa values** have greatest CA inhibitory and diuretic activity.
- The benzene meta sulphonamide derivatives have activity only when substituted with **chlorine** or **methyl groups**.

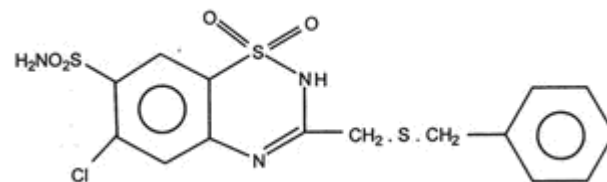
II. Thiazide and Thiazide like Diuretics (Site-III Diuretics)



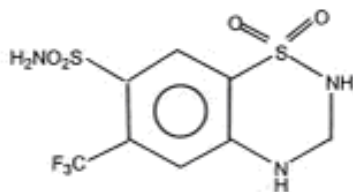
chlorthiazide



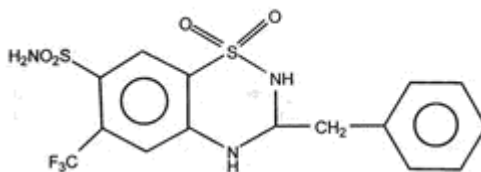
Hydrochlorthiazide



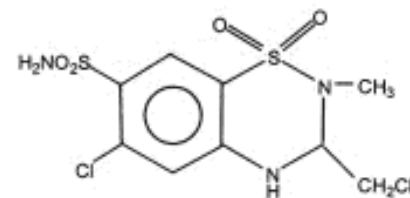
Benzthiazide



Hydroflumethiazide



Bendroflumethiazide



Methylchlorothiazide

DIURETICS...

Structure Activity Relationship for Thiazides

- **Thiazides** having **benzothiadiazine 1,1-dioxide** with weakly acidic character is important for good activity.
- **Presence of electron withdrawing group at C-6** is necessity for good diuretic activity. Substitution of chlorine at C-6 has good activity.
- Substitution of **CF₃** group has more lipid soluble and larger diuretic action than Chloro compound.
- **Presence of electron donating groups like methyl or methoxy at C-6** reduces the diuretic activity.

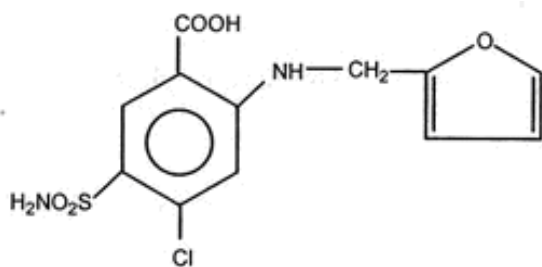
DIURETICS...

- Removal or replacement of **sulphonamide** at C-7 reduces the diuretic activity.
- **Saturation of double bond** between 3&4 having 10 times more diuretic activity than unsaturated analogue.
- Introduction of lipophilic groups such as **aralkyl, halo alkyl, thioether** enhances the **diuretic activity** and increase the **duration of action**.
- Alkyl substitution at N2 lowers the polarity and enhances the duration of action.

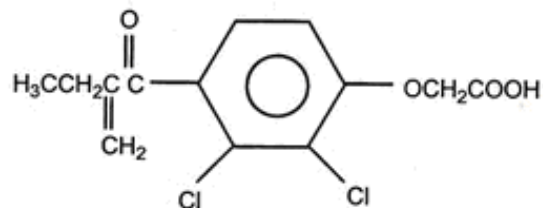
Mechanism of action of Thiazides

- These drugs blocks the reabsorption of Na^+ , Cl^- exchange in the **distal convoluted tubule** by inhibiting the luminal membrane-bound **Na^+/Cl^- co transport system**.

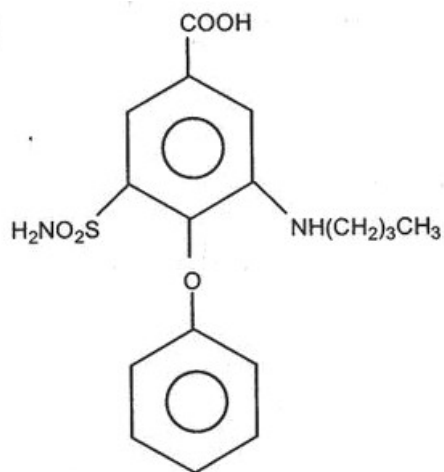
III. Loop Diuretics (Site-II Diuretics)



Furosemide



Ethacrynic acid



Bumetanide

By Yitagesu T.(MSc.)

DIURETICS...

Structure Activity Relationship for Loop diuretics

- 5-sulfomoyl and 2-aminobenzoic acid group is required for good diuretic activity.
- Substitution at 1st position must be acidic for good diuretic activity.
- The activating group at 4th position can be Cl or CF₃ group increases the activity.
 - Phenoxy, alkoxy, anilino, benzyl or benzoyl groups substituted at 4th position decreases diuretic activity.
- Furfuryl, benzyl and thienyl methyl group at 2-position increases the activity.

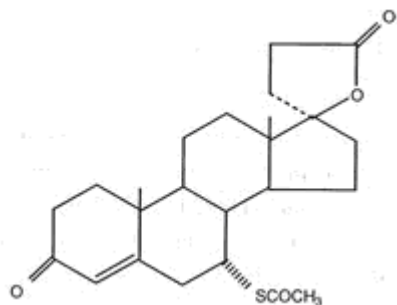
DIURETICS...

Mechanism of action of Loop Diuretics

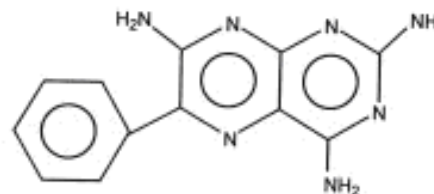
- The diuretics inhibit the $\text{Na}^+/\text{K}^+/\text{Cl}^-$ co-transport system located in the luminal membrane of cells in the limb of Henle's loop.
- The carboxylate moiety is responsible for their competing with Cl^- for the Cl^- binding site on $\text{Na}^+/\text{K}^+/\text{Cl}^-$ co-transport system.

DIURETICS...

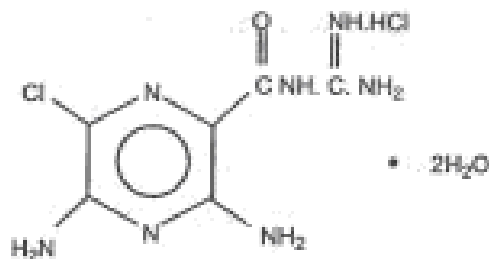
IV. Potassium sparing Diuretics (Site-IV Diuretics)



Spironolactone (aldactone)



Triamterene



Amiloride hydrochloride

By Yitagesu T.(MSc.)

DIURETICS...

- **Spiranolactone** inhibits the reabsorption of 2-3% of the filtered load sodium at site-IV by competitively inhibiting the action of aldosterones.
- **Triamterene & Amiloride** decrease the antiluminal membrane bound $\text{Na}^+/\text{K}^+-\text{ATP}$ ase activity, leads to decrease in cellular extusion of sodium and in the cellular uptake of potassium at site-IV.